

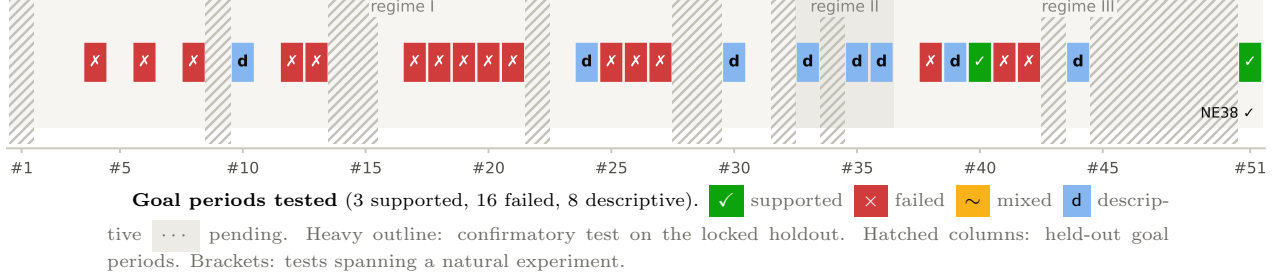
H126: Goal occupancy is a telegraph process: dwell times predict the occupancy

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. An assigned goal acts on both telegraph rates: it raises the on-rate and lowers the off-rate ($\Delta \ln k_{\text{off}} < 0$ in 15/17 kickoffs and in the #12 debate schedule), so it is a symmetric field, not an attempt-rate field. Dwell shape is not identifiable here.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH367 models each agent's on-goal state as a random telegraph signal. In that model dwells are exponential, two mean dwells predict the occupancy $p = k_{\text{on}} / (k_{\text{on}} + k_{\text{off}})$, and a kickoff raises only k_{on} . H126 fits a two-state hidden Markov model to H105's decoy-threshold on-goal labels on each agent's ledger call clock: 39 non-holdout units, 17 kickoffs, the #12 debate schedule, #51 private goals and Opus 5's reassignment. The decisive result is that a kickoff lowers k_{off} as well as raising k_{on} : $\Delta \ln k_{\text{off}}$ is below 0 in 15 of 17 kickoffs. The dwell-shape and occupancy clauses are not identifiable at village sampling (synthetic, before data).

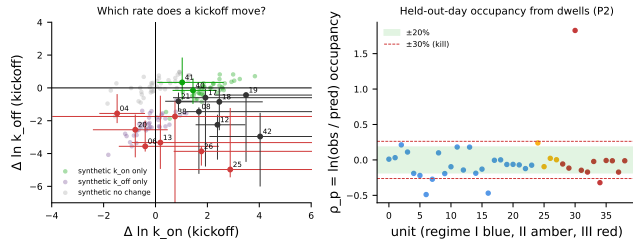
PHYSICAL MODEL

Telegraph spin read through a noisy channel (models 10 and 02). The latent $s_{i,c} \in \{\text{off}, \text{on}\}$ of agent i switches per ledger call with $a_i = k_{\text{on}}$ and $b_i = k_{\text{off}}$. Statements are observed at irregular call gaps Δ and labelled on-goal with sensitivity q_1 and false-positive rate q_0 . Rival: a 4-state aggregated chain whose dwells are mixtures of two geometrics. Kickoff designs fit segment rates before and after the boundary with shared emissions.

$$P^\Delta = \Pi + \lambda^\Delta (I - \Pi), \quad \lambda = 1 - a - b, \quad \Pi_{\cdot, \text{on}} = \frac{a}{a+b}$$

$$P(b_k=1 | s) = q_0 + (q_1 - q_0) \mathbb{I}[s = \text{on}], \quad \tau_{\text{on}} = 1/b, \quad \tau_{\text{off}} = 1/a$$

one-rate field: $\Delta \ln a > 0$, $\Delta \ln b = 0$; Glauber: $\Delta \ln a = -\Delta \ln b$



Observables. Left: kickoff rate changes with 90% agent-bootstrap CIs (green: k_{on} only; red: k_{off} dominant; black: both move). Faint points are synthetic k_{on} -only, k_{off} -only and no-change worlds on the real skeletons. Most real kickoffs sit below zero on the k_{off} axis. Right: held-out-day occupancy ratio ρ_p per unit; the $\pm 30\%$ kill fires in 5/39 units. The synthetic shows this check passes under every rival.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, a goal works by holding agents on goal as much as by recruiting them. A kickoff multiplies the per-call chance of switching onto the goal by about 5 and divides the chance of leaving it by about 6 (medians over 17 kickoffs). Latent on-goal stints last a median of 78 calls. Expect occupancy to respond to anything that interrupts agents (lower k_{off} is what a goal buys), not only to how often it is announced.

RESULTS VS PREDICTIONS

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

Prediction (dated)	Observed	Verdict
P1 exponential dwells	shape test power 0.05–0.10 (synthetic); “heavy” 6/39	not identifiable
P2 p from dwells within 20%	28/39; kill ($> 30\%$) 5/39; passes in all synthetic worlds	non-discriminating
P3 kickoff raises k_{on} only	2/17; k_{off} falls in 15/17	failed
P4 occupancy carried by k_{on}	$\rho(\ln k_{\text{on}}, \text{logit } p)$ 0.37; $\rho(\ln k_{\text{off}})$ −0.50	failed
P5 call clock beats wall (III)	9/11 units	supported
N1 #12 debate field on k_{on} only	+1.27 / −1.37 [−2.28, −0.35]	failed
N2 #51 per-agent rates persist	median ρ 0.61 (8 pairs)	supported
N3 NE38 k_{on} only	+3.2 / −5.1 [−12.9, +0.2]	weak pass
Overall (A1 rule)	P3 fails	failed

CAVEATS

The on-goal label is a text-embedding proxy (bge decoy threshold); gte doubles the heavy-tail counts. In each kickoff’s pre-segment, occupancy along the next goal sits at the decoy floor ($p \approx 0.04$), so its k_{off} rests on few latent stints, although the synthetic gave 0/30 false k_{off} calls under a true k_{on} -only change. Sparse regime-III units give unstable short dwells. A kickoff smoke test printed one result (K38) after Amendment A1 and before the full run.

ROUND 2

Calibrate the classifier externally (Jev on-goal sample) to restore shape power; fit semi-Markov dwell hazards on #51 own-goal sequences; fit time-dependent rates through the kickoff day; run the frozen confirm (held-out kickoffs, C1: k_{off} falls in $\geq 2/3$).

- **H126-R1** A calibrated classifier.
- **H126-R2** A semi-Markov fit for #51.
- **H126-R3** Kickoff time course.

GOAL PERIODS IN DETAIL

G04 × failed
G06 × failed
G08 × failed
G10 d descriptive
G12 × failed
G13 × failed
G17 × failed
G18 × failed
G19 × failed
G20 × failed
+17 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

H126-telegraph-goal-occupancy · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H126-telegraph-goal-occupancy/README.md · ratings: H126 agent (suggested) (page 2)